IEORE4004: Optimization Models & Methods **Fall 2025**

Complementary Slackness And Applications

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1 Introduction

We have seen in previous lectures that the solutions of primal and dual LP problems are closely related. We now investigate this fact more closely by introducing some conditions that an optimal pair of primal and dual solutions have to satisfy. Those conditions are called *complementary slackness*. We will then see that complementary slackness can be used to enrich our understanding of the dual of certain LPs, as well as find solutions to some interesting problems.

2 Complementary Slackness

Consider the following primal and dual pairs

$$\begin{array}{lrcl} \max c^T x & & & \\ \text{s.t.} & & & \\ (P) & Ax & \leq & b \\ & x & \geq & 0 \end{array} \qquad and \qquad \begin{array}{lrcl} \min b^T y & & & \\ \text{s.t.} & & & \\ (D) & A^T y & \geq & c \\ & y & \geq & 0 \end{array}$$

and let $\bar{x}$ and $\bar{y}$ be optimal solutions for the primal and the dual, respectively. We have

$$c^T \bar{x} \leq (A^T \bar{y})^T \bar{x} = \bar{y}^T A \bar{x} \leq \bar{y}^T b = c^T \bar{x}, \tag{1}$$

where all inequalities and equations are deduced similarly to the proof of weak duality, while the last equation follows from strong duality. If we focus on the left-most and right-most term in (1), we have

$$c^T \bar{x} \leq ... \leq c^T \bar{x},$$

which implies that all inequalities in (1) must be satisfied at equality. Thus, we must have

$$c^T \bar{x} = (A^T \bar{y})^T \bar{x} \tag{2}$$

and

$$\bar{y}^T A \bar{x} = \bar{y}^T b. \tag{3}$$

(2) can be written as

$$\sum_i c_i x_i = \sum_i [A^T \bar{y}]_i \bar{x}_i,$$

where $[A^T \bar{y}]_i$ is the i-th component of the vector $A^T \bar{y}$. Since $[A^T \bar{y}]_i \geq c_i$ by dual feasibility, we must have

$$[A^T \bar{y}]_i \bar{x}_i = c_i x_i \tag{4}$$

for all i. For this to happen, for every i we must have:

- Either $[A^T\bar{y}]_i = c_i$,
- or $\bar{x}_i = 0$ (or both).

Similarly, from (3), we deduce that for every j, we have:

- Either $[A\bar{x}]_j = b_j$,
- or $\bar{y}_j = 0$ (or both).

We can summarize this observation in the following theorem.

Theorem 1. *Let $\bar{x}$, $\bar{y}$ be candidate solutions for (P) and (D) above. Then $\bar{x}$ is optimal for the primal and $\bar{y}$ is optimal for the dual if and only if the following happens:*

1. *$\bar{x}$ if feasible for the primal (P), and*
2. *$\bar{y}$ is feasible for the dual (D), and*
3. *$\bar{x}$ and $\bar{y}$ are* complementary, *i.e.:*
   - *If $\bar{x}_i > 0$ for some i, then the corresponding dual constraint is satisfied at equality by $\bar{y}$.*
   - *If $\bar{y}_i > 0$ for some i, then the corresponding primal constraint is satisfied at equality by $\bar{x}$.*

Note that the negation of "$\bar{x}$ is optimal for the primal and $\bar{y}$ is optimal for the dual" is "at least one of $\bar{x}$ and $\bar{y}$ is not optimal". In particular, one of the two could still be optimal.

2.1 For generic Linear Programs

Similarly to what we did for duality theory, we can extend complementary slackness to LPs in general form. Recall the conversion table between primal and dual:

Primal (P') (Max)	Dual (D') (Min)
i-th constraint $\leq$	variable $y_i \geq 0$
i-th constraint $\geq$	variable $y_i \leq 0$
i-th constraint $=$	variable y_i unrestricted (free)
$x_i \geq 0$	i-th constraint $\geq$
$x_i \leq 0$	i-th constraint $\leq$
x_i unrestricted (free)	i-th constraint $=$

For such general problems, Theorem 1 has the following form.

Theorem 2. *Let $\bar{x}$, $\bar{y}$ be candidate solutions for (P') and (D') above. Then $\bar{x}$ is optimal for the primal and $\bar{y}$ is optimal for the dual if and only if the following happens:*

1. *$\bar{x}$ if feasible for the primal (P'), and*
2. *$\bar{y}$ is feasible for the dual (D'), and*
3. *$\bar{x}$ and $\bar{y}$ are* complementary, *i.e.:*
   - *If x_i is not free and $\bar{x}_i \neq 0$ for some i, then the corresponding dual constraint is satisfied at equality by $\bar{y}$.*
   - *If y_i is not free and $\bar{y}_i \neq 0$ for some i, then the corresponding primal constraint is satisfied at equality by $\bar{x}$.*

2.2 An Example

Consider the primal problem

$$\begin{aligned} \max \quad & z = 3x_1 + x_2 - x_3 \\ \text{s.t.} \quad & 2x_1 + x_2 - x_3 \le 4 \qquad (1) \\ & -x_1 + 2x_2 + x_3 = 1 \qquad (2) \\ & x_1 \ge 0, \quad x_2 \text{ free}, \quad x_3 \le 0. \end{aligned}$$

and its Dual

$$\begin{aligned} \min \quad & w = 4y_1 + y_2 \\ \text{s.t.} \quad & 2y_1 - y_2 \ge 3 \qquad (x_1) \\ & y_1 + 2y_2 = 1 \qquad (x_2) \\ & -y_1 + y_2 \le -1 \qquad (x_3) \\ & y_1 \ge 0, \qquad y_2 \text{ free}. \end{aligned}$$

We next show how to deduce optimal solutions for both the primal and the dual using complementary slackness. Note that this is not always possible, in the sense that we cannot always find a dual optimal and deduce from it From $y_1 + 2y_2 = 1$ we have $y_1 = 1 - 2y_2$. Substituting into the other dual constraints:

$$\begin{aligned} 2(1-2y_2) - y_2 \ge 3 \quad &\Longleftrightarrow \quad -5y_2 \ge 1 \iff y_2 \le -\tfrac{1}{5}, \\ -(1-2y_2) + y_2 \le -1 \quad &\Longleftrightarrow \quad -1 + y_2 \le -1 \iff y_2 \le 0. \end{aligned}$$

Thus the most stringent constraint it $y_2 \le -\frac{1}{5}$. Note that we can replace $y_1 = 1 - 2y_2$ in the objective function of the dual, as to obtain

$$w = 4y_1 + y_2 = 4(1 - 2y_2) + y_2 = 4 - 7y_2,$$

so w is minimized by the largest feasible y_2, namely $y_2^* = -\frac{1}{5}$. Hence

$$y_1^* = 1 - 2\Big(-\tfrac{1}{5}\Big) = \tfrac{7}{5} = 1.4, \qquad y_2^* = -\tfrac{1}{5} = -0.2, \qquad w^* = 4y_1^* + y_2^* = \tfrac{27}{5} = 5.4.$$

We now use complementary slackness to recover a primal optimum.

- Since $y_1^* > 0$, primal constraint (1) is tight:

  $$2x_1 + x_2 - x_3 = 4.$$

- Dual constraint for x_3 at (y_1^*, y_2^*) is slack:

  $$-y_1^* + y_2^* = -\tfrac{7}{5} - \tfrac{1}{5} = -\tfrac{8}{5} < -1,$$

  hence by CS, $x_3^* = 0$ (recall $x_3 \le 0$).

- Using (2) and $x_3^* = 0$:

  $$\begin{cases} 2x_1 + x_2 = 4, \\ -x_1 + 2x_2 = 1, \end{cases} \quad \Longrightarrow \quad x_2^* = \tfrac{6}{5} = 1.2, \qquad x_1^* = \tfrac{14}{10} = 1.4.$$

Objective value:
$$z^* = 3x_1^* + x_2^* - x_3^* = 3\left(\tfrac{7}{5}\right) + \tfrac{6}{5} - 0 = \tfrac{27}{5} = 5.4 = w^*,$$
so $(x_1^*, x_2^*, x_3^*) = (1.4, 1.2, 0)$ and $(y_1^*, y_2^*) = (1.4, -0.2)$ are optimal.

Now recall the optimal pair found earlier:
$$x^* = \left(\tfrac{7}{5},\, \tfrac{6}{5},\, 0\right), \qquad y^* = \left(\tfrac{7}{5},\, -\tfrac{1}{5}\right).$$

Let us now use the optimal solution to deduce that two other given solutions are non-optimal. Let us start with $\hat{x} = (1, 1, 0)$ is non-optimal.

Since $y_1^* = \frac{7}{5} > 0$ (not free), so primal constraint (1) must be tight at an optimal x:
$$2x_1 + x_2 - x_3 = 4.$$

But for $\hat{x} = (1, 1, 0)$,
$$2(1) + 1 - 0 = 3 \;\neq\; 4,$$
so the CS condition with y^* fails. Hence $\hat{x}$ cannot be optimal.

Similarly, let $\hat{y} = (3, -1)$.

Here $x_1^* = \frac{7}{5} > 0$ with sign $x_1 \geq 0$ (not free), so the dual constraint for x_1 must be tight at an optimal y:
$$2y_1 - y_2 = 3.$$

But for $\hat{y} = (3, -1)$,
$$2(3) - (-1) = 7 \;\neq\; 3,$$
so the CS condition with x^* fails. Hence $\hat{y}$ cannot be optimal.

3 Applications

3.1 The production problem

Recall the production problem where we want to decide how many tables (x_1) and chairs (x_2) we want to produce as to maximize our profit, subject to the limits on the amount of wood (first constraint) and paint (second constraint) we have available.

$$\begin{array}{rrcrcl} \max & 3x_1 & + & 2x_2 & & \\ & x_1 & + & x_2 & \leq & 80 \\ & 2x_1 & + & x_2 & \leq & 100 \\ & & & x_1, x_2 & \geq & 0. \end{array} \qquad \text{(P)}$$

Recall that its dual

$$\begin{array}{rrcrcl} \min & 80y_1 & + & 100y_2 & & \\ & y_1 & + & 2y_2 & \geq & 3 \\ & y_1 & + & y_2 & \geq & 2 \\ & & & y_1, y_2 & \geq & 0 \end{array} \qquad \text{(D)}$$

can be interpreted as the problem “solved by the market” to determine the price of wood (y_1) and paint (y_2) that guarantee that the production company has no interest in buying or selling wood and paint.

Let $\bar{x}, \bar{y}$ be candidate primal and dual pairs, respectively. By Theorem 1, Complementary slackness on x_1 is as follows:

$$\bar{x}_1 > 0 \;\Rightarrow\; \bar{y}_1 + 2\bar{y}_2 = 3.$$

How can we interpret this relation? If in the optimal production plan $\bar{x}$ we decide to build some tables, then the total amount of money we would make by selling the wood and paint needed to produce the table (this is $\bar{y}_1 + 2\bar{y}_2$) equals the profit of using them to produce and sell the table.

Complementary slackness on y_1 is as follows:

$$\bar{y}_1 > 0 \;\Rightarrow\; \bar{x}_1 + \bar{x}_2 = 80.$$

Hence, if we could sell wood at a strictly positive price on the market, then in the optimum production plan we will use all the wood (else, we could sell the leftover and make more money that we are making).

Similar deductions apply to x_2 and y_2 and, more generally, to problems with multiple production options and resource constraints.

3.2 The diet problem

Consider the following problem

- There are m nutrients and n foods;
- For our diet, we want at least $b_i > 0$ units of nutrient i ($i = 1, \ldots, m$);
- Each unit of food j ($j = 1, \ldots, n$) costs $c_j > 0$ and contains $a_{i,j} \geq 0$ units of nutrient i ($i = 1, \ldots, m$).

Our goal is to find the quantities of food so that we take enough of each nutrient, at minimum cost. A linear program for this problem can be written as follows.

$$\begin{array}{rrcl} \min & \sum_{j=1}^{n} c_j x_j & & \\ \text{(P)} & \sum_{j=1}^{n} a_{i,j} x_j & \geq & b_i,\ i = 1, \ldots, m \\ & x & \geq & 0 \end{array}$$

and its dual is

$$\begin{array}{rrcl} \max & \sum_{i=1}^{m} b_i y_i & & \\ \text{(D)} & \sum_{i=1}^{m} a_{i,j} y_i & \leq & c_j, \quad j = 1, \ldots, n \\ & y & \geq & 0 \end{array}$$

Again, the dual has an economic interpretation as the "pill seller problem": a company sells m pills, each one containing a unit of a specific nutrient i, and want to decide their prices $(y_1, \ldots, y_m)$ so that they maximize their profit, subject to the constraint that the amount of pills that replace a certain type of food should not cost more than the food itself. The dual constraints

We now give an interpretation of the complementary slackness conditions:

- $x_j > 0 \Rightarrow \sum_i a_{i,j} y_j = c_j$: if we decide to buy food j, then the cost of the pills that can supply the same amount of nutrients as food j is equal to the cost of food j (else, we would not buy food j and replace it instead with the appropriate pills);
- $y_i > 0 \Rightarrow \sum_j a_{i,j} x_j = b_i$ or, equivalently, $\sum_j a_{i,j} x_j > b_i \Rightarrow y_i = 0$: if the diet provides strictly more than the needed amount of nutrient i, then there is no point in buying the corresponding pill, whose price should therefore be equal to 0.

3.2.1 An example

Consider the following numerical instance of the diet problem with $m = 2$ nutrients and $n = 3$ foods, and requirements / nutrient content as in the LP below.

$$\begin{aligned} \min\ & 4x_1 + 5x_2 + 6x_3 \\ \text{s.t. } & 3x_1 + x_2 + 2x_3 \geq 8, \\ & x_1 + 3x_2 + 2x_3 \geq 6, \\ & x \geq 0. \end{aligned}$$

Its dual is

$$\begin{aligned} \max\ & 8y_1 + 6y_2 \\ \text{s.t. } & 3y_1 + y_2 \leq 4, \\ & y_1 + 3y_2 \leq 5, \\ & 2y_1 + 2y_2 \leq 6, \\ & y \geq 0. \end{aligned}$$

Consider the two solutions for the primal:

$$\hat{x} = (1, 0, 2.5), \qquad \bar{x} = (9/4, 5/4, 0),$$

and the two solutions for the dual:

$$\hat{y} = (1, 1), \qquad \bar{y} = (7/8, 11/8).$$

We want to check their optimality using complementary slackness. We see that $\bar{x}$ is feasible for the primal, and $\bar{y}$ is feasible for the dual. As for the complementarity properties,

- $\bar{x}_1 > 0$ implies that the first dual constraint must be satisfied at equality by $\bar{y}$ (which holds true).
- $\bar{x}_2 > 0$ implies that the second dual constraint must be satisfied at equality by $\bar{y}$ (which holds true).
- $\bar{x}_3 = 0$, no restriction on the third dual constraint.
- $\bar{y}_1 > 0$ implies that the first primal constraint must be satisfied at equality by $\bar{x}$ (which holds true).
- $\bar{y}_2 > 0$ implies that the second primal constraint must be satisfied at equality by $\bar{x}$ (which holds true).

We conclude that $\bar{x}$ is optimal for the primal and $\bar{y}$ is optimal for the dual. As for an economic interpretation, since we do not buy any unit of food 3 ($\bar{x}_3 = 0$), then the pill providing the same nutrient as food 3 can cost less of the price of food 3 (the 3rd dual constraint is not tight at $\bar{y}$).

We leave as an exercise to use complementary slackness to check that $\hat{x}, \hat{y}$ are not optimal.

3.3 The assignment game

In the *assignment game*, we consider a market where

- $b_1, \ldots, b_n$ are buyers willing to buy a house;
- $s_1, \ldots, s_m$ are sellers, each of them wants to sell their house;
- For all i, j, buyer b_i has a *valuation* $v_{ij} \geq 0$ for the house of seller s_j (think of the valuation as the money the buyer thinks the house is worth).

Thus, an instance of the assignment game can be seen as a graph[1] with two set of nodes, one representing buyers and the other representing sellers. There is an edge between b_i and s_j if $v_{ij} > 0$. A graph where we can partition the set of nodes into two sets, say A and B, so that there is no edge between nodes in A, and similarly there is no edge between nodes in B, is called *bipartite*. Note that the graph we just constructed is bipartite.

Example. An illustrative instance (with two buyers and three houses) is given by the valuation matrix

$$V = \left(v_{ij}\right) = \begin{bmatrix} v_{11} & v_{12} & v_{13} \\ v_{21} & v_{22} & v_{23} \end{bmatrix} = \begin{bmatrix} 6 & 3 & 5 \\ 4 & 7 & 2 \end{bmatrix}.$$

Here, for example, $v_{11} = 6$ is the value buyer b_1 assigns to house s_1, and $v_{22} = 7$ is the value buyer b_2 assigns to house s_2.

Our goal is to decide a fair way to choose who buys what and at which price. Given a graph $G(V, E)$, a *matching* is a set $M \subseteq E$ such that each $v \in V$ is contained in at most one edge from M. With the graph notation, a matching can be used to denote who is buying which house (i.e., b_i buys the house of s_j if and only if the edge $b_i s_j$ belongs to the matching M).

Our main solution concept will be that of *equilibrium*, which is a pair (M, p) defined as follows:

- M is a matching;
- $p \geq 0$ is a vector of prices of houses;
- (Unsold house cannot be discounted further) If s_j does not sell their house (i.e., there is no edge of M that contains s_j), then $p_j = 0$;
- (Buyers won't budge) Each b_i is assigned a house j that maximizes their utility at the given prices: $j \in \arg\max_\ell u_i(\ell, p)$.

In the definition above, the utility of buyer b_i for the house of seller s_j at prices p is defined as $u_i(j, p) = v_{ij} - p_j$. It is therefore given by the valuation the buyer has for the house, minus its price.

An equilibrium for the example above. In the example above, take the matching

$$M = \{\, b_1 s_1,\ b_2 s_2 \,\}$$

(seller s_3 remains unmatched), and the price vector

$$\bar{p} = (\bar{p}_1, \bar{p}_2, \bar{p}_3) = (0.5, 3.4, 0).$$

Then $\bar{p} \geq 0$ and the unsold house s_3 has $\bar{p}_3 = 0$. Buyer utilities at these prices are

$$\begin{aligned} &u_1(1, \bar{p}) = 6 - 0.5 = 5.5, \quad u_1(2, \bar{p}) = 3 - 3.4 = -0.4, \quad u_1(3, \bar{p}) = 5 - 0 = 5, \\ &u_2(1, \bar{p}) = 4 - 0.5 = 3.5, \quad u_2(2, \bar{p}) = 7 - 3.4 = 3.6, \qquad u_2(3, \bar{p}) = 2 - 0 = 2. \end{aligned}$$

Thus

$$u_1(1, \bar{p}) = 5.5 = \max\{u_1(1, \bar{p}), u_1(2, \bar{p}), u_1(3, \bar{p})\} \quad \text{and} \quad u_2(2, \bar{p}) = 3.6 = \max\{u_2(1, \bar{p}), u_2(2, \bar{p}), u_2(3, \bar{p})\},$$

so each buyer is assigned a house that maximizes their utility at prices $\bar{p}$. Therefore, $(M, \bar{p})$ is an equilibrium for this market.

[1] A graph is a network where arcs are not oriented, and are called edges. See the lecture notes on flow problems.

A detour: Linear Programming for matching problems. An important step in our procedure to compute an equilibrium is given by an LP for solving the following *maximum weighted matching problem*:

Given: A bipartite graph $G(V, E)$ with $V = A \cup B$ and a valuation function $v : E \to \mathbb{Z}$ on the edges;

Find: a matching M of G that maximizes $v(M) = \sum_{ij \in M} v(ij)$.

An LP that solves the previous problem is given by the following:

$$(P)\quad \begin{array}{rll} \max & \sum_{i,j} v_{i,j}\, x_{i,j} & \\ \text{s.t.} & \sum_{j \in B} x_{i,j} \leq 1 & \text{for all } i \in A \\ & \sum_{i \in A} x_{i,j} \leq 1 & \text{for all } j \in B \\ & x \geq 0. & \end{array}$$

Here, x_{ij} is interpreted as a variable that is going to have value 1 if edge ij belongs to the matching and 0 otherwise. In particular, there will always be an optimal solution whose variables assume values in $\{0, 1\}$ (so either 0, or 1, but no fractional value)[2] and which therefore corresponds to a matching. The first set of constraint says that, for every node i of A, at most one edge of the matching can contain i. The second set of constraints says that, for every node j of B, at most one edge of the matching can contain j. The dual of (P) is

$$\begin{array}{rll} \min & \sum_i u_i + \sum_j p_j & \\ \text{s.t.} & u_i + p_j \geq v_{i,j} & \text{for all } i \in A, j \in B \quad (D) \\ & u, p \geq 0. & \end{array}$$

Continuing the example above. The Linear Program for computing a matching maximizing the valuations in the example above is: Primal:

$$\begin{array}{rll} \max & 6x_{11} + 3x_{12} + 5x_{13} + 4x_{21} + 7x_{22} + 2x_{23} & \\ \text{s.t.} & x_{11} + x_{12} + x_{13} \leq 1 & (\text{buyer } b_1), \\ & x_{21} + x_{22} + x_{23} \leq 1 & (\text{buyer } b_2), \\ & x_{11} + x_{21} \leq 1 & (\text{seller } s_1), \\ & x_{12} + x_{22} \leq 1 & (\text{seller } s_2), \\ & x_{13} + x_{23} \leq 1 & (\text{seller } s_3), \\ & x_{ij} \geq 0 & (i = 1, 2;\ j = 1, 2, 3). \end{array} \tag{P$_{\text{ex}}$}$$

Dual:

$$\begin{array}{rl} \min & u_1 + u_2 + p_1 + p_2 + p_3 \\ \text{s.t.} & u_1 + p_1 \geq 6, \quad u_1 + p_2 \geq 3, \quad u_1 + p_3 \geq 5, \\ & u_2 + p_1 \geq 4, \quad u_2 + p_2 \geq 7, \quad u_2 + p_3 \geq 2, \\ & u_1, u_2, p_1, p_2, p_3 \geq 0. \end{array} \tag{D$_{\text{ex}}$}$$

The matching given before corresponds to the point $\bar{x}$ with $x_{11} = 1, x_{22} = 1$, and all other $\bar{x}_{ij} = 0$. The price vector $\bar{p}$ given before is $\bar{p}_1 = 0.5$, $\bar{p}_2 = 3.4$, $\bar{p}_3 = 0$. Using e.g. complementary slackness, one verifies that they are optimal primal and dual solutions, respectively. As we see next, this is not by chance!

Back to the assignment game. Let us now see how linear programming and complementary slackness can be used to compute an equilibrium. First, write the LP (P) for the *maximum weighted matching problem* on the graph defined

[2] Note that this fact requires a proof, that we are omitting.

by the assignment game instance, where weights are given by valuations, and its dual (D). Consider $\bar{x}$ optimal for (P) and $\binom{\bar{u}}{\bar{p}}$ optimal for (D). We already argued that $\bar{x}$ corresponds to a matching. Moreover, by Complementary slackness we have:

1. $\bar{p} \geq 0$ (by dual feasibility);
2. $\bar{u}_i \geq v_{i,j} - \bar{p}_j \quad$ for all i, j (by dual feasibility);
3. $\bar{x}_{i,j} = 1 \Rightarrow \bar{u}_i = \bar{v}_{i,j} - \bar{p}_j$ (by complementary slackness);
4. $\bar{p}_j > 0 \Rightarrow \sum_i \bar{x}_{i,j} = 1$ (by complementary slackness).

We claim that $(\bar{x}, \bar{p})$ is an equilibrium and $\bar{u}$ is the utility of buyers. In fact, we already showed that $\bar{x}$ is a matching. 1. above implies nonnegativity of prices. 2. above and 3. above imply that a buyer is assigned a house that, at the given price, maximizes their utility. The contrapositive of 4. above implies that, if a house is not sold, then its price must be 0. Thus, $(\bar{x}, \bar{p})$ is an equilibrium.

So from optimal primal and dual solutions to the LPs, we can find an equilibrium for the Assignment Game.